

Apibunya "Alpha" Sinworn

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Builds at the intersection of hardware, software, and art. Designs and ships complete experiences end to end, from interactive installations and live event technology to full-stack AI products. Always the art student in the engineering room, and the engineering student in the art room.

EXPERIENCE

IT Specialist and Software Developer | Thai Living Law, Bangkok 2026-Present

- Built and deployed the firm's full content automation platform end to end as an active commercial tool
- Integrated OpenRouter, Meta Graph API, LINE Messaging API, and Sanity CMS into a single publishing dashboard

Product Owner Intern | AIYA, Bangkok September 2025

- Researched the competitive landscape and defined product requirements for an AI software-as-a-service platform
- Developed a software interface connecting AI hardware to LINE OA and customer relationship management platforms

Instructional Aide and Teaching Assistant | Arizona State University 2022-2024

- Designed curriculum and ran labs as Microcontroller Applications Instructional Aide, troubleshooting Arduino builds
- Led recitation sessions and created supplementary materials as Signals and Systems Teaching Assistant
- Taught the human-centered design process and monitored 70+ student projects as Engineering Projects in Community Service Teaching Assistant

SELECTED WORK

The Sparkling Hour: Year 2126 | what a WHY x NEXTOPIA, Siam Paragon, Bangkok 2026

- Built a scroll-driven cinematic hero website in Next.js, React Three Fiber, and GSAP, using canvas image-sequence scrubbing of AI-generated video
- Directed and programmatically assembled a promo reel remixing Fritz Lang's Metropolis (1927) through an ffmpeg pipeline with speech-to-text caption alignment
- Shipped a live event activity where attendee submissions appeared in real time on a projected 3D globe, each triggering AI sticker generation via Supabase Realtime and Gemini
- Deployed the full stack across Netlify and Supabase, configuring reverse-proxy routing so both apps served under a single event domain

AI Content Automation Platform | Thai Living Law, Bangkok 2026

- Designed a bilingual Thai and English content pipeline carrying legal content from AI topic discovery through drafting, lawyer approval, and multi-channel publishing
- Built a no-login shareable approval flow so lawyers could review and sign off from their phones
- Published to Facebook, Instagram, LINE OA, and the firm's website from one dashboard

SUKONTHnAI, AI Scent Installation | Bangkok February 2026

- Created an interactive installation that captures a visitor's photo, reads its color, mood, and energy with AI, and dispenses a custom Thai herbal fragrance in roughly 20 seconds
- Architected a Raspberry Pi 5 and Pico system communicating over MQTT through Node-RED, mapping Gemini Flash 2.0 image analysis to 26 essential oils across 4 scent profiles
- Conceived, engineered, and built solo in approximately 52 hours

PKratheoy and KaDealDed | CodeCamp Thailand 2025

- Built PKratheoy, a mental health journaling app with a mood calendar and cognitive behavioral therapy tools
- Built KaDealDed, a deals aggregation platform with authentication, QR codes, and an admin dashboard, serving as Product Owner

NASA L'SPACE Mission Concept Academy | Project Manager Summer 2024

- Led a 10-person cross-timezone team designing a lunar rover mission concept
- Managed schedules, milestones, and cross-disciplinary collaboration across time zones

NASA L'SPACE Proposal Writing and Evaluation Experience Academy Spring 2024

- Trained in NASA proposal evaluation and review alongside subject matter experts
- Developed working knowledge of NASA project evaluation criteria

Head Movement Cursor Control, Assistive Technology | Arizona State University Spring 2024

- Built a cursor control system for people with disabilities using a head-mounted accelerometer and NI LabVIEW
- Designed the hardware-software interface and tuned sensor algorithms for usability and responsiveness

LeafyWear, Wearable Peritoneal Dialysis Device | ASU Capstone 2023-2024

- Led electrical design for circuit, motor control, and power management on a device freeing patients from 4-hour clinic sessions
- Conducted stakeholder research with patients, nurses, and physicians to drive user-centered design
- Won 1st Place Popular Vote at Open Pitch and presented at the ASU Biomedical Engineering Symposium

Optogenetic PACE Machine V3, Honors Thesis | ASU Bartelle Lab 2023-2024

- Designed and fabricated a custom printed circuit board and motor shield for a 5-pump open-source bioreactor used in immunotherapy nanobody research

- Iterated the full cycle from breadboard to protoboard to fabricated board, moving from Fusion360 to Altium
- Programmed Arduino automation for media feed control and growth tracking

PCB Silkscreen Artwork, PACE Machine V3 | ASU Bartelle Lab

2023-2024

- Drew original artwork in Adobe Illustrator and embedded it directly into the silkscreen layer of a functional circuit board
- Merged art practice and engineering fabrication into a single object, making a research-grade instrument the canvas

EPICS Lake Litter Solution | Arizona State University

2023-2024

- Built an autonomous water robot to collect trash at Tempe Town Lake as Electrical and Finance Lead
- Led the pitching team at EPICS Elite Pitch, taking 3rd Place

Space Cow Apocalypse, Board Game | Arizona State University

Spring 2023

- Directed all game pieces, packaging, and visual identity as Art Lead using Adobe Illustrator, Photoshop, laser cutting, and 3D printing
- Grounded mechanics in user research and statistical analysis of playtesting data

EPICS Coral Reef Restoration, Study Abroad Spain | Arizona State University

Fall 2022

- Led a team of 7 designing artificial coral reefs for field testing in Spain
- Applied engineering to environmental restoration across a cross-cultural collaboration

EPICS Mental Health Mobile Games | Arizona State University

Fall 2021

- Designed mental health app interface and experience for the Hopi community in collaboration with Northern Arizona University
- Created culturally responsive layout and graphics grounded in community input

EDUCATION

Intelligent Automation Certificate | KMITL, Bangkok

2026

Programmable Logic Controller programming (Mitsubishi/GX Works), ladder logic, industrial automation, and ABB robot programming with path control and sorting on a physical robot arm

Raspberry Pi and IoT | TESR Academy, Bangkok

January 2026

ROS 2 Basic | TESR Academy, Bangkok

November 2025

Full-Stack Developer Trainee | CodeCamp Thailand, Bangkok

2025

JavaScript, React, Node.js, Express, MySQL, and Git

B.S.E. Biomedical Engineering | Arizona State University, Ira A. Fulton Schools of Engineering

2019-2024

Barrett, the Honors College | Magna Cum Laude | Dean's List | GPA 3.68/4.00

Honors Thesis: Open Source Hardware for Accessible Directed Evolution Bioreactor

High School Diploma | Triamudom Suksa, Bangkok

2018

COMMUNITY

Global Shapers Community, World Economic Forum, Bangkok Hub

2025-Present

- Led InterGen Harmony, a youth-elder mental health initiative supported by the Bangkok Metropolitan Administration and the Thai Health Promotion Foundation
- Presented project outcomes to policymakers at parliament

RECOGNITION

- Top 40 Finalist, OKMD Hackathon (2025)
- Finalist, MIT Hacking Medicine x Siriraj (2024)
- 1st Place Popular Vote, Open Pitch, Wearable Dialysis Device, Arizona State University (2024)
- 3rd Place, EPICS Elite Pitch, Arizona State University (2024)

SKILLS

Creative Technology: React Three Fiber, WebGL, GSAP, generative AI video and image pipelines, ffmpeg, interactive installation, live event technology

Software: TypeScript, JavaScript, React, Next.js, Python, C/C++, Node.js, Node-RED, N8N, MQTT, Supabase, PostgreSQL, Docker, MATLAB, LabVIEW

Hardware: Raspberry Pi, Arduino, printed circuit board design (Altium, Fusion 360), sensors and actuators, motor control, soldering, 3D printing, Programmable Logic Controller (Mitsubishi), ABB robot arm

Design: Adobe Illustrator, Photoshop, Figma, SOLIDWORKS

AI-augmented development: rapid prototyping and hardware-software integration using AI tools

Languages: Thai (Native), English (Fluent)